Temporal Gains, Spatial Costs: Revisiting Video Fine-Tuning in Multimodal Large Language Models

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Research Questions

- How does Video-SFT (video-based supervised fine-tuning) affect the balance between spatial and temporal understanding capabilities in MLLMs?
- What is the relationship between temporal budget (number of frames) and the image-video performance trade-off?
- Can adaptive frame allocation strategies mitigate the conflict between image and video understanding in unified MLLM training?

Core Idea

This paper reveals a fundamental conflict in multimodal large language models: video fine-tuning consistently improves video understanding but degrades image understanding, termed the "temporal trap". The key insight is that shared parameters create competing gradients - video-oriented updates with temporal specialization interfere with spatial capability preservation. The authors show this trade-off intensifies with more training frames and propose an adaptive frame allocation strategy that assigns different frame counts based on instruction complexity to partially mitigate the conflict.

Key Formula

Image loss change under video update

$$L_{img}(\theta^{+}) = L_{img}(\theta) - \eta \langle g_{img}, g_{vid}^{(m)} \rangle + \frac{\eta^2}{2} (g_{vid}^{(m)})^{\top} H_{img}(\tilde{\theta}_{img}) g_{vid}^{(m)}$$

Shows how image loss L_img changes when parameters θ are updated by video gradient g^(m)_vid. The key term ⟨g_img, g^(m)_vid⟩ represents gradient alignment - negative alignment leads to image degradation. This formalizes when video updates harm image performance.

Video gradient decomposition

$$g_{vid}^{(m)} = g_{sh} + \alpha^{(m)} g_{tmp} + \epsilon^{(m)}$$

Decomposes video gradient into shared component g_sh (useful for both modalities), temporal component g_tmp (video-specific), and residual ε^(m). Coefficient α^(m) increases with frame count m, explaining why more frames intensify the conflict.

Key Figure

SFT stage are expected. However, our results reveal a different phenomenon: MLLMs exhibit a conflict between image and video modalities. After Video-SFT, video understanding improves while image understanding degrades. We refer to this phenomenon as the temporal trap.

3.2 Study Dimensions

We conduct systematic experiments along three key dimensions.

- ♣ **Model architecture**: Representative MLLMs including Qwen2.5-VL, LLaVA-Next-Video, and LLaVA-1.5.
- ♦ **Model scale**: Four scales of Qwen2.5-VL with 3B, 7B, 32B, and 72B parameters.
- ♠ **Frame sampling setting**: Videos are uniformly sampled with 8, 16, 32, and 64 frames during Video-SFT.

3.3 Datasets and Evaluation

Based on the LLaVA-Next-Video-178k (Zhang et al., 2024b) dataset, we curate a training dataset of 20,000 videos collected from 10 different sources, covering diverse instruction formats such as textual descriptions, open-ended questions, and multiple-choice questions to ensure substantial data diversity. Training data statistics are reported in the supplementary material.

Evaluation datasets are selected from commonly used benchmarks for MLLMs. The image benchmarks include MME (Fu et al., 2023), MMStar (Chen et al., 2024), MMBench (Liu et al., 2024a), and POPE (Li et al., 2023), while the video benchmarks consist of Video-MME (Fu et al., 2025), MVBench (Li et al., 2024b), TempCompass (Liu et al., 2024b), and Video-MMMU (Hu et al., 2025). These benchmarks cover the core visual abilities, including coarse- and fine-grained perception, cognition, and hallucination.

4 The Temporal Trap behind Visual Modality Conflict

Although videos are composed of sequences of images and share the same visual encoder, improvements in video understanding do not transfer to static image understanding. We observe a systematic trade-off: Video-SFT enhances video performance while often degrading image performance. We refer to this phenomenon as the temporal trap,

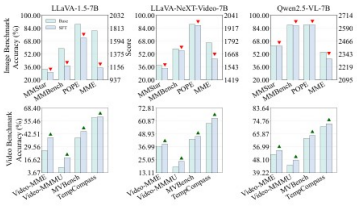

Figure 2: Comparison of image and video benchmark performance before and after Video-SFT across different MLLMs. ▼ denotes performance degradation after SFT, whereas ▲ indicates performance improvement.

which reflects an intrinsic conflict between temporal adaptation and spatial visual reasoning. To better understand this phenomenon, we analyze it along three key dimensions: model architecture (Sec. 4.1), model scale (Sec. 4.2), and fine-tuning frame count (Sec. 4.3).

4.1 Impact of Model Architecture

Figure 2 shows a consistent trend in all architectures evaluated. After Video-SFT, the performance on the video benchmarks improves for nearly every model, while most image benchmarks exhibit clear degradation. This pattern reveals a clear conflict between image and video modalities: although Video-SFT improves video understanding, it simultaneously weakens static image reasoning.

The magnitude of the conflict varies across architectures. LLaVA-1.5 exhibits the largest performance drop in the image benchmarks, while LLaVA-NeXT-Video shows a smaller gap. Qwen2.5-VL remains comparatively stable, indicating that stronger spatial–temporal alignment and

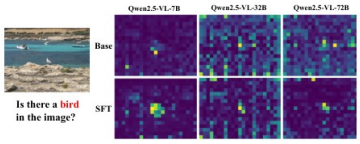

Figure 3: **Cross-scale attention visualizations before and after Video-SFT on Qwen2.5-VL models (7B, 32B, 72B).** For the query "Is there a bird in the image?", attention becomes more dispersed in smaller models after Video-SFT, while larger models retain more localized focus on the target object, suggesting improved robustness to the temporal trap.

Figure 2: Shows consistent performance degradation on image benchmarks (MME, MMStar, MMBench, POPE) while video benchmarks improve across different MLLM architectures after Video-SFT, directly demonstrating the temporal trap phenomenon.

Limitations

- Hybrid-Frame Strategy relies on predefined discrete frame counts (8, 16, 32, 64) rather than continuous optimization

- Current approach only uses textual instruction analysis, ignoring actual visual content dynamics of videos
- Theoretical analysis provides local sufficient conditions but not complete characterization of the trade-off mechanism
- Evaluation limited to standard benchmarks, excluding streaming video and interactive scenarios